



2025

O23(B)

BOOKLET NO.

221176

**Geology**  
**Paper – II**

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

**Question Paper Specific Instructions**

*Please read each of the following instructions carefully before attempting questions :*

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

**Note :** Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

**SECTION – A**

- Q1.** Answer the following questions in about **150** words. **(5×10=50)**
- Q1. (a)** What are crystal forms and its types ? Discuss in brief closed crystal form. **10**
- Q1. (b)** Discuss about assimilation of magma. **10**
- Q1. (c)** Describe the mineral assemblages and rocks in zeolite facies related to protolith composition. **10**
- Q1. (d)** Write in brief about the chemical stability of common heavy minerals, give its significance in provenance studies. **10**
- Q1. (e)** What are ino-silicates ? Discuss the subtle difference between single chain and double chain ino-silicates along with their structural characteristics. **10**
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- Q2. (a)** Elucidate with suitable examples the concept of crystal symmetry and different symmetry elements found in crystals. Add a note on how these symmetry elements are used to classify crystals into various systems. **20**
- Q2. (b)** Explain the relationship between crystal planes and crystallographic axes with representative crystal faces that cut one, two or three crystallographic axes. **15**
- Q2. (c)** What are granite ? Classify granite on the basis of mineralogy and petrogenesis. **15**
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- Q3. (a)** Discuss the factors controlling metamorphism and the resulting metamorphic minerals. Add a note on graphical representation of mineral paragenesis. **20**
- Q3. (b)** Explain metamorphic zone. Describe the various metamorphic zones based on the index mineral isogrades. **15**
- Q3. (c)** Enumerate the diagnostic features of glacial environment. Add a note on glacial deposit. **15**
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- Q4. (a)** What are physical properties of minerals ? How properties like hardness, cleavage, luster and specific gravity are related to atomic structure and bonding within minerals ? **20**
- Q4. (b)** How orthoclase is distinguished from plagioclase feldspar ? Add a note on their physical and optical properties. **15**
- Q4. (c)** Elucidate the optical properties of common rock forming minerals viz. pleochroism, extinction angle, double refraction, birefringence, twinning and dispersion. **15**
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**SECTION – B**

- Q5.** Answer the following questions in about **150** words. **(5×10=50)**
- Q5. (a)** Give a brief account on Manganese nodules. **10**
- Q5. (b)** Explain geometric method of ore reserve calculation. **10**
- Q5. (c)** Describe polymorphism crystal with suitable examples. **10**
- Q5. (d)** Explain the impact of sea level changes that occurred during the Quaternary period. **10**
- Q5. (e)** Describe the significance of ore, ore minerals and gangue minerals in calculating tenor of ore. Add a note on magmatic process of ore formation. **10**
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- Q6. (a)** Discuss the concept of ore localisation in primary and secondary environment. **20**
- Q6. (b)** What are the essential criteria and industrial uses of barytes that classify it as an industrial mineral ? **15**
- Q6. (c)** What is flotation ? Give its principles, reagent used and steps involved in carrying out the flotation process. **15**
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- Q7. (a)** Elucidate the geochemical classification, migration and distribution of elements in the Earth. Add a note on role of crystal chemistry and trace elements in geochemical processes. **20**
- Q7. (b)** Discuss the composition of Earth's mantle. **15**
- Q7. (c)** Describe the various types of floods. Discuss flood hazards in India and suggest the management strategy to mitigation them. **15**
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- Q8. (a)** Discuss the textural features of ores and their significance in understanding the genesis of ore deposits. **20**
- Q8. (b)** Elaborate the National Mineral Policy and its significance in Country's development. **15**
- Q8. (c)** Describe the geological setting, mode of occurrence, mineralogy and geographical distribution of uranium and thorium. **15**
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